Lecture 15: Wave Packets and Free Dispersion

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Lecture 14 introduced the attractive delta-function well and found the decaying bound-state solution away from the origin. In this lecture we complete that calculation, distinguish plane waves from localized states, construct a Gaussian wave packet, and study its free spreading and motion.

Questions.

- How does the derivative jump determine the bound state of the delta well?
- Why are plane waves useful even though they are not normalizable on the full line?
- How are the position and momentum widths of a Gaussian packet related?
- Why does a free Gaussian packet spread?
- What is the dispersion time, and what does it measure?
- How does the center of a moving Gaussian packet evolve?

Lecture roadmap. We first finish the delta-well bound state by applying the derivative jump condition and normalization. We then move from bound states to free-particle plane waves and localized Gaussian packets. Finally, we derive the position and momentum uncertainties, the free-particle spreading law, and the trajectory of the packet center.

Review from Lecture 14

For the attractive delta-function well,

$$V(x) = -g\delta(x), \qquad g > 0, \tag{1}$$

the stationary Schrödinger equation is

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} - g\delta(x)\right]\phi(x) = E\phi(x). \tag{2}$$

A bound state lies below the outside potential, so $E < 0$. Defining

$$\alpha = \frac{\sqrt{-2mE}}{\hbar} > 0, \tag{3}$$

normalizability and continuity give

$$\boxed{\phi(x) = Ae^{-\alpha|x|}.} \tag{4}$$

The remaining task is to determine α, E, and A.

1 Completing the delta-well bound state

1.1 Derivative jump at the origin

Because the delta potential is singular, $\phi'(x)$ need not be continuous at the origin. Integrating the Schrödinger equation from $-\varepsilon$ to $+\varepsilon$ gives

$$-\frac{\hbar^2}{2m}\int_{-\varepsilon}^{\varepsilon}\phi''(x)\,\mathrm{d}x - g\int_{-\varepsilon}^{\varepsilon}\delta(x)\phi(x)\,\mathrm{d}x = E\int_{-\varepsilon}^{\varepsilon}\phi(x)\,\mathrm{d}x. \tag{5}$$

As $\varepsilon \to 0$, the right-hand side vanishes, while the delta integral selects $\phi(0)$. Therefore

$$\boxed{\phi'(0^+) - \phi'(0^-) = -\frac{2mg}{\hbar^2}\phi(0).} \tag{6}$$

For $\phi = Ae^{-\alpha|x|}$,

$$\phi'(0^+) = -\alpha A, \qquad \phi'(0^-) = +\alpha A. \tag{7}$$

Substitution into the jump condition gives

$$-2\alpha A = -\frac{2mg}{\hbar^2}A, \qquad \boxed{\alpha = \frac{mg}{\hbar^2}.} \tag{8}$$

The bound-state energy is consequently

$$\boxed{E = -\frac{\hbar^2\alpha^2}{2m} = -\frac{mg^2}{2\hbar^2}.} \tag{9}$$

1.2 Normalization and physical meaning

Normalization determines the remaining constant:

$$1 = \int_{-\infty}^{\infty} |\phi(x)|^2 \, \mathrm{d}x \tag{10}$$

$$= 2|A|^2 \int_0^{\infty} e^{-2\alpha x} \, \mathrm{d}x = \frac{|A|^2}{\alpha}. \tag{11}$$

Choosing $A > 0$, the normalized bound state is

$$\boxed{\phi(x) = \sqrt{\alpha}\, e^{-\alpha|x|}, \qquad \alpha = \frac{mg}{\hbar^2}.} \tag{12}$$

The jump condition selects exactly one positive value of α, so it selects exactly one bound-state energy. Thus

$$\boxed{\text{The attractive delta well has exactly one bound state.}} \tag{13}$$

Increasing g increases α, shortens the decay length $1/\alpha$, and makes the energy more negative relative to the outside threshold.

2 From bound states to scattering states

Bound states are localized and have discrete energies. Positive-energy free states can propagate to infinity and form a continuous spectrum. They are the building blocks used to describe incident and outgoing particles in scattering:

$$\boxed{\text{incoming state} \longrightarrow V(x) \longrightarrow \text{outgoing state.}} \tag{14}$$

Here *scattering* has a broad meaning: light reaching the eye after interacting with an object and a particle beam passing through matter are both scattering processes. The central question is how a potential transforms an incident wave.

	Bound state	**Scattering state**
spatial behavior	decays as $\lvert x\rvert \to \infty$	extends to infinity
energy spectrum	discrete	continuous
normalization	$\int \lvert\phi\rvert^2 \, \mathrm{d}x = 1$	Dirac-delta normalization

3 Free-particle plane waves

For $V(x) = 0$, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\phi''(x) = E\phi(x). \tag{15}$$

For $E > 0$, its general solution is

$$\boxed{\phi_E(x) = Ae^{ikx} + Be^{-ikx}, \qquad E = \frac{\hbar^2 k^2}{2m}.} \tag{16}$$

Including the time-dependent phase gives

$$e^{i(kx-\omega t)}: \qquad p = +\hbar k, \qquad \text{right-moving}, \tag{17}$$

$$e^{-i(kx+\omega t)}: \qquad p = -\hbar k, \qquad \text{left-moving}, \tag{18}$$

where

$$\boxed{\omega(k) = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}.} \tag{19}$$

For a single plane wave,

$$|e^{ikx}|^2 = 1, \qquad \int_{-\infty}^{\infty} |e^{ikx}|^2 \, \mathrm{d}x = \infty. \tag{20}$$

Therefore a plane wave is not a normalizable localized state on the full line. It is a generalized eigenfunction with definite momentum. A localized physical state is instead formed by superposing plane waves:

$$\boxed{\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widetilde{\psi}(k,0) e^{ikx - i\omega(k)t} \, \mathrm{d}k.} \tag{21}$$

	Plane wave	**Wave packet**
momentum	definite $p = \hbar k$	range of momenta
position	extends over all space	localized
normalization	not normalizable	normalizable
uncertainty	$\Delta p = 0$	$\Delta p > 0$

Questions.

Question 1. If a plane wave is not normalizable on the full line, why is it still useful?

Answer and explanation. A plane wave has a definite momentum and is a generalized eigenfunction of the free-particle Hamiltonian. Plane waves form a continuous basis, so a normalizable localized state can be constructed as their superposition. The coefficient $\widetilde{\psi}(k,0)$ specifies the momentum content of that state.

4 A free Gaussian wave packet

4.1 Initial state and position uncertainty

Consider the normalized initial state

$$\boxed{\psi(x,0) = \frac{1}{(\pi a^2)^{1/4}} \exp\!\left(-\frac{x^2}{2a^2}\right).} \tag{22}$$

Its probability density is

$$|\psi(x,0)|^2 = \frac{1}{\sqrt{\pi}\,a} e^{-x^2/a^2}. \tag{23}$$

The density is even, so

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\psi(x,0)|^2 \,\mathrm{d}x = 0. \tag{24}$$

Using the Gaussian integral,

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 |\psi(x,0)|^2 \,\mathrm{d}x = \frac{a^2}{2}. \tag{25}$$

Therefore

$$\boxed{\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{a}{\sqrt{2}}.} \tag{26}$$

The parameter a controls the width, but it is not itself the standard deviation; the standard deviation is $a/\sqrt{2}$.

4.2 Momentum distribution and uncertainty

Using the Fourier convention

$$\psi(x,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widetilde{\psi}(k,0) e^{ikx} \,\mathrm{d}k, \tag{27}$$

the momentum-space wave function is

$$\boxed{\widetilde{\psi}(k,0) = \left(\frac{a^2}{\pi}\right)^{1/4} e^{-a^2k^2/2}.} \tag{28}$$

Thus

$$|\widetilde{\psi}(k,0)|^2 = \frac{a}{\sqrt{\pi}} e^{-a^2k^2}, \qquad \langle k \rangle = 0, \qquad \langle k^2 \rangle = \frac{1}{2a^2}. \tag{29}$$

It follows that

$$\boxed{\Delta k = \frac{1}{\sqrt{2}a}, \qquad \Delta p = \hbar \Delta k = \frac{\hbar}{\sqrt{2}a}.} \tag{30}$$

Consequently, this Gaussian is a minimum-uncertainty state:

$$\boxed{\Delta x \, \Delta p = \frac{\hbar}{2}.} \tag{31}$$

A smaller a produces stronger position localization but a broader momentum distribution.

Questions.

Question 2. If the initial position width of the Gaussian packet is reduced, what happens to Δp?

Answer and explanation. Since

$$\Delta x = \frac{a}{\sqrt{2}}, \qquad \Delta p = \frac{\hbar}{\sqrt{2}a},$$

reducing a decreases Δx but increases Δp. Their product remains $\hbar/2$.

4.3 Energy content

The momentum distribution is centered at $k_0 = 0$. The energy of its central plane-wave component is therefore

$$\boxed{E_0 \equiv E(k_0) = \frac{\hbar^2 k_0^2}{2m} = 0.} \tag{32}$$

This does not mean that the whole packet has zero energy. Since localization requires a nonzero momentum spread,

$$\langle H \rangle = \frac{\hbar^2}{2m}\langle k^2 \rangle = \boxed{\frac{\hbar^2}{4ma^2}} > 0. \tag{33}$$

For the Gaussian, $\langle k^4 \rangle = 3/(4a^4)$. Hence

$$\Delta E = \sqrt{\langle H^2 \rangle - \langle H \rangle^2} \tag{34}$$

$$= \boxed{\frac{\hbar^2}{2\sqrt{2}\, ma^2}}. \tag{35}$$

Thus E_0, $\langle H \rangle$, and ΔE describe different things:
- E_0 is the energy of the central plane-wave component;
- $\langle H \rangle$ is the mean result of many energy measurements;
- ΔE measures the spread of possible energy results.

5 Free time evolution and spreading

Each plane-wave component evolves with its own phase:

$$e^{ikx} \longrightarrow e^{ikx - i\omega(k)t}, \qquad \omega(k) = \frac{\hbar k^2}{2m}. \tag{36}$$

Therefore the freely evolving packet is

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widetilde{\psi}(k,0) e^{ikx - i\hbar k^2 t/(2m)}\, \mathrm{d}k. \tag{37}$$

Using

$$\int_{-\infty}^{\infty} e^{-Ak^2 + Bk}\, \mathrm{d}k = \sqrt{\frac{\pi}{A}} e^{B^2/(4A)}, \qquad \operatorname{Re} A > 0, \tag{38}$$

gives

$$\boxed{\psi(x,t) = \frac{1}{(\pi a^2)^{1/4}} \frac{1}{\sqrt{1 + i\hbar t/(ma^2)}} \exp\left[-\frac{x^2}{2a^2[1 + i\hbar t/(ma^2)]}\right].} \tag{39}$$

Taking the modulus squared gives a Gaussian density at every time:

$$\boxed{|\psi(x,t)|^2 = \frac{1}{\sqrt{\pi}\, a(t)} \exp\left[-\frac{x^2}{a^2(t)}\right],} \tag{40}$$

where

$$\boxed{a(t) = a\sqrt{1 + \left(\frac{\hbar t}{ma^2}\right)^2}, \qquad \Delta x(t) = \frac{a(t)}{\sqrt{2}}.} \tag{41}$$

The integral of the density remains one, but its peak decreases as its width grows. This is free-particle dispersion.

5.1 Dispersion time

Define the characteristic dispersion time

$$\boxed{\tau = \frac{ma^2}{\hbar}.} \tag{42}$$

The width can then be written as

$$a(t) = a\sqrt{1 + (t/\tau)^2}. \tag{43}$$

Its limiting behavior is

time scale	behavior
$\lvert t\rvert \ll \tau$	$a(t) \approx a$ (little spreading)
$\lvert t\rvert = \tau$	$a(t) = \sqrt{2}\, a$
$\lvert t\rvert \gg \tau$	$a(t) \approx \dfrac{\hbar\lvert t\rvert}{ma}$

(44)

Thus τ is the time at which the width parameter has grown by a factor of $\sqrt{2}$. A narrower initial packet has a smaller τ and spreads more rapidly.

5.2 What the simulation shows

For a packet with zero central momentum, the simulation illustrates three results:

- The center remains at its initial position while the packet spreads according to $a(t) = a\sqrt{1 + (t/\tau)^2}$.
- Decreasing the initial width increases the momentum and energy spreads and decreases τ, so the packet spreads faster.
- Although the central component has $E_0 = 0$, the localized packet has $\langle H\rangle > 0$ and $\Delta E > 0$.

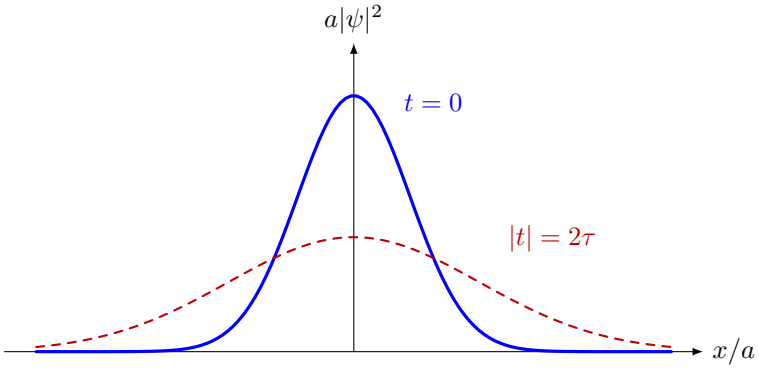


Figure 1: A freely evolving Gaussian remains normalized while its peak falls and its width increases.

6 A moving Gaussian

6.1 Initial state and energy content

Translate the packet to x_0 and give it a central wave number k_0:

$$\boxed{\psi(x,0) = \frac{1}{(\pi a^2)^{1/4}} e^{-(x-x_0)^2/(2a^2)} e^{ik_0 x}.} \tag{45}$$

The Gaussian envelope determines the initial position, while the phase factor $e^{ik_0 x}$ supplies the mean momentum:

$$\boxed{\langle x\rangle_0 = x_0, \qquad \langle p\rangle_0 = p_0 = \hbar k_0.} \tag{46}$$

More explicitly, its momentum-space wave function is

$$\widetilde{\psi}(k,0) = \left(\frac{a^2}{\pi}\right)^{1/4} \exp\left[-\frac{a^2(k-k_0)^2}{2}\right] e^{-i(k-k_0)x_0},$$

and hence

$$|\widetilde{\psi}(k,0)|^2 = \frac{a}{\sqrt{\pi}} e^{-a^2(k-k_0)^2}.$$

Compared with the non-moving Gaussian, the distribution has only been shifted from $k = 0$ to $k = k_0$; its shape and width are unchanged. Therefore

$$\langle k\rangle = k_0, \qquad \left\langle (k-k_0)^2 \right\rangle = \frac{1}{2a^2},$$

so

$$\boxed{\langle p\rangle = p_0 = \hbar k_0, \qquad \Delta p = \hbar \Delta k = \frac{\hbar}{\sqrt{2}\, a}.}$$

The factor involving x_0 is a pure phase in momentum space and therefore does not affect the momentum probability distribution.

The energy of the central component is

$$\boxed{E_0 = \frac{p_0^2}{2m} = \frac{\hbar^2 k_0^2}{2m}.} \tag{47}$$

For the complete packet,

$$(\Delta p)^2 = \langle p^2\rangle - \langle p\rangle^2 \quad \Longrightarrow \quad \langle p^2\rangle = \langle p\rangle^2 + (\Delta p)^2.$$

Using

$$\langle p\rangle = p_0, \qquad \Delta p = \frac{\hbar}{\sqrt{2}\,a},$$

we obtain

$$\begin{aligned}\langle H\rangle &= \frac{\langle p^2\rangle}{2m} = \frac{\langle p\rangle^2 + (\Delta p)^2}{2m} \qquad (48)\\ &= \frac{p_0^2}{2m} + \frac{1}{2m}\left(\frac{\hbar}{\sqrt{2}\,a}\right)^2\\ &= \boxed{E_0 + \frac{\hbar^2}{4ma^2}}. \qquad (49)\end{aligned}$$

The first term is the energy of the central momentum component. The second is the additional mean kinetic energy associated with the packet's momentum spread. Thus the mean energy exceeds E_0 because a localized packet contains a range of momenta.

6.2 Probability density and packet center

We now evolve the momentum-space amplitude. Each component acquires the phase

$$e^{-i\hbar k^2 t/(2m)}.$$

Now set $q = k - k_0$. Since

$$k^2 = k_0^2 + 2k_0 q + q^2,$$

the Fourier integral becomes

$$\psi(x,t) = \frac{e^{ik_0 x - i\hbar k_0^2 t/(2m)}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{a^2}{\pi}\right)^{1/4} \exp\left[-\frac{a^2 q^2}{2} - \frac{i\hbar q^2 t}{2m}\right] \times \exp\left[iq\left(x - x_0 - \frac{\hbar k_0}{m}t\right)\right] \mathrm{d}q.$$

This is the same Gaussian integral as for the non-moving packet, with x replaced by

$$X = x - x_0 - \frac{\hbar k_0}{m}t = x - x_0 - \frac{p_0}{m}t.$$

Therefore

$$\psi(x,t) = \frac{1}{(\pi a^2)^{1/4}} \frac{e^{ik_0 x - i\hbar k_0^2 t/(2m)}}{\sqrt{1 + it/\tau}} \exp\left[-\frac{\left(x - x_0 - \frac{p_0}{m}t\right)^2}{2a^2(1 + it/\tau)}\right].$$

The plane-wave factor has unit modulus. Also,

$$\left|\frac{1}{\sqrt{1 + it/\tau}}\right|^2 = \frac{1}{\sqrt{1 + (t/\tau)^2}}, \qquad \mathrm{Re}\left(\frac{1}{1 + it/\tau}\right) = \frac{1}{1 + (t/\tau)^2}.$$

Taking the modulus squared thus gives

$$\boxed{|\psi(x,t)|^2 = \frac{1}{\sqrt{\pi}\,a(t)} \exp\left[-\frac{\left(x - x_0 - \frac{p_0}{m}t\right)^2}{a^2(t)}\right],} \tag{50}$$

with the same spreading width

$$a(t) = a\sqrt{1 + (t/\tau)^2}. \tag{51}$$

To calculate its expectation value, define

$$y = x - x_0 - \frac{p_0}{m}t, \qquad x = y + x_0 + \frac{p_0}{m}t. \tag{52}$$

Then

$$\langle x\rangle_t = \int_{-\infty}^{\infty} x|\psi(x,t)|^2\,\mathrm{d}x \tag{53}$$

$$= \int_{-\infty}^{\infty} \left(y + x_0 + \frac{p_0}{m}t\right) \frac{e^{-y^2/a^2(t)}}{\sqrt{\pi}\,a(t)}\,\mathrm{d}y \tag{54}$$

$$= x_0 + \frac{p_0}{m}t. \tag{55}$$

The term proportional to y integrates to zero because it is odd, and the remaining Gaussian integral equals one by normalization. We therefore define the packet center by

$$\boxed{x_c(t) \equiv \langle x\rangle_t = x_0 + \frac{p_0}{m}t = x_0 + \frac{\hbar k_0}{m}t.} \tag{56}$$

The packet both moves and spreads: $x_c(t)$ describes the motion of its center, whereas $a(t)$ describes the growth of its width.

Lecture 15 summary

- The derivative jump condition completes the attractive delta-well solution:

$$\phi(x) = \sqrt{\alpha}\, e^{-\alpha|x|}, \qquad \alpha = \frac{mg}{\hbar^2}, \qquad E = -\frac{mg^2}{2\hbar^2}.$$

- A plane wave has definite momentum but is not localized or normalizable on the full line. A localized wave packet is a superposition of plane waves.
- For the initial Gaussian,

$$\Delta x = \frac{a}{\sqrt{2}}, \qquad \Delta p = \frac{\hbar}{\sqrt{2}a}, \qquad \Delta x\, \Delta p = \frac{\hbar}{2}.$$

- A free Gaussian remains Gaussian but spreads:

$$a(t) = a\sqrt{1 + \left(\frac{\hbar t}{ma^2}\right)^2}, \qquad \tau = \frac{ma^2}{\hbar}.$$

- A moving Gaussian has separate center and width dynamics:

$$x_c(t) = \langle x \rangle_t = x_0 + \frac{p_0}{m}t, \qquad \Delta x(t) = \frac{a(t)}{\sqrt{2}}.$$

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., the sections on the delta-function potential, free-particle wave packets, Fourier transforms, and Gaussian spreading. Review the distinction among a plane wave, a stationary scattering eigenfunction, and a normalizable localized wave packet.